

Anthocyanins are bioavailable in humans following an acute dose of cranberry juice.

Research suggests that anthocyanins from berry fruit may affect a variety of physiological responses, including endothelial function, but little information is available regarding the pharmacokinetics of these flavonoids in humans. To determine the pharmacokinetics of cranberry anthocyanins, a study was undertaken in 15 participants (age: 62 $\pm$ 8 y) with coronary artery disease. Blood and urine samples were collected between baseline (0 h) and 4 h after consumption of 480 mL cranberry juice (54% juice; 835 mg total polyphenols; 94.47 mg anthocyanins). Marked inter-individual differences in plasma anthocyanin pharmacokinetics were observed with maximum anthocyanin concentrations detected between 1 and 3 h. Cranberry anthocyanins were bioavailable but with notable differences in the maximum concentration and area under the curve(0-4h) between individual participants. The pattern of anthocyanin glucosides observed in plasma and urine generally reflected the relative concentration determined in the juice. Plasma concentrations of the individual anthocyanins ranged between 0.56 and 4.64 nmol/L. Total recovery of urinary anthocyanin was 0.79 $\pm$ 0.90% of the dose delivered. These data are in agreement with the pharmacokinetics of anthocyanins from other foods suggesting that cranberry anthocyanins are poorly absorbed and rapidly removed from plasma. Observed concentrations of plasma anthocyanins appear insufficient to alter radical load or redox potential but may be adequate to affect signal transduction and/or gene expression.